

Tingting Du

UW-Madison | tdu35@wisc.edu | GitHub | LinkedIn | Google Scholar | Website

Interest: Robotics; Human-Robot Collaboration; Vision-Language-Action Models; LLM Reasoning.



EDUCATION

University of Wisconsin-Madison

Madison, Wisconsin, USA

B.S. in Computer Science and Mathematics;

Jan. 2025 ~ May. 2027

- GPA: 3.86/4.00; Dean's List; Autonomous Robotics, Operating Systems, Neural Networks.

University of California, Berkeley

Berkeley, California, USA

Visiting Student in Computer Science;

Aug. 2023 ~ Jun. 2024

- GPA: 3.77/4.00; Global Access Scholarship; Real Analysis, Data Structures, Algorithms, AI.

Ningbo University

Ningbo, Zhejiang, China

Undergraduate in Linguistics;

Sep. 2021 ~ Jun. 2023

- GPA: 3.80/4.00; Zhejiang Provincial Scholarship (Top 3%, 2×).

RESEARCH

Trimanual Manipulation

RT² Lab, University of Wisconsin-Madison

Advisor: Prof. Mike Hagenow

Aug. 2026 ~ Present

- Learning from demonstration for trimanual human-robot team manipulation;
- Wired a unified emergency-stop interlock tying all three arms to a single safety circuit.

Vision-Language-Action and World Action Models

CASE Lab, University of Maryland

Advisor: Prof. Ang Li

Nov. 2025 ~ Aug. 2026

- Multi-layer shared projector bridging 2D/3D representations in VLA models (NeurIPS 2026);
- Geometry-enhanced world action model (Under Review);
- Deployed and evaluated both on real-world manipulation with a UFACTORY xArm.

Student Modeling for Question Generation

DM² Lab, University of Notre Dame

Advisor: Prof. Meng Jiang

Jul. 2024 ~ Jan. 2025

- Developed a student modeling approach for question generation (ACL 2025).

Situated Language Use and Convention Formation

BAIR, University of California, Berkeley

Advisor: Prof. Alane Suhr

Feb. 2024 ~ Jun. 2024

- Analyzed ad-hoc convention formation in a large 3D-environment dialogue corpus.

PUBLICATIONS

- T. Du, Z. Wang, G. Sun, A. Li, "XGenAct: Geometry-Enhanced World Action Models through Cross-Task Generation," arXiv:2610.03516, 2026.
- G. Sun, T. Du, K. Feng, C. Luo, Z. Shen, et al., "ROCKET: Residual-Oriented Multi-Layer Alignment for Spatially-Aware VLA Models," NeurIPS, 2026.
- Z. Wang, B. Wang, H. Zhang, T. Du, T. Chen, et al., "Vision-Language-Action in Robotics: A Survey of Datasets, Benchmarks, and Data Engines," TMLR, 2026.
- X. Tang, T. Qin, T. Peng, Z. Zhou, D. Shao, T. Du, et al., "Agent KB: A Hierarchical Memory Framework for Cross-Domain Agentic Problem Solving," ICML Workshop on Collaborative and Federated Agentic Workflows, 2025.
- B. Nguyen, T. Du, M. Yu, L. Angrave, M. Jiang, "QG-SMS: Enhancing Test Item Analysis via Student Modeling and Simulation," ACL, 2025.
- N. Tomlin, N. Zhou, E. Fleisig, L. Chen, T. Wright, L. Vinh, L.X. Ma, S. Eisape, T. Du, et al., "Characterizing Language Use in a Collaborative Situated Game," arXiv:2512.03381, 2025.

SKILLS

- Robotics: ROS 2, UFACTORY xArm, multi-arm workcell bring-up and safety integration, Onshape CAD, 3D printing.
- Machine Learning: PyTorch, HuggingFace Transformers, vLLM, PEFT/LoRA, RLBench.
- Languages & Infra: Python, C/C++, Java, Julia, R, JS/TS, SQL, Assembly, L^AT_EX; Linux, Docker, AWS.